FORM 2
THE PATENTS
ACT 1970 (39 of
1970)
&
THE PATENTS (AMENDMENT)
RULES, 2006 COMPLETE
SPECIFICATION
(See section 10 and rule 13)

1. TITLE OF THE INVENTION:

A CLOUD-BASED INTELLIGENT IRRIGATION DECISION SYSTEM WITH UNCERTAINTY-AWARE CONTROL AND UTILITY POWER FALLBACK

2. APPLICANTS:

Bannari Amman Institute of Technology, Sathyamangalam. Erode, Tamil Nadu, India– 638401.

3. PREAMBLE TO THE DESCRIPTION

COMPLETE

The following specification particularly describes the invention and the manner in which it is to be performed.

4. DESCRIPTION

FIELD OF THE INVENTION

[01] The present invention relates to smart agriculture, automated irrigation control systems, and decision-support platforms. More particularly, the invention relates to a cloud-based intelligent irrigation decision system incorporating environmental sensing, uncertainty-aware weather analysis, adaptive energy management, and utility power fallback for efficient irrigation control.

BACKGROUND OF THE INVENTION

[02] Efficient irrigation management is critical for sustainable agriculture, especially in regions facing water scarcity and unpredictable rainfall patterns. Conventional irrigation systems either rely on manual operation or simple threshold-based automation, which often leads to over-irrigation or under-irrigation.

[03] Existing smart irrigation systems utilise soil moisture sensors and weather data; however, such systems generally lack mechanisms to handle uncertainty in weather forecasts. Rain predictions may be inaccurate, resulting in improper irrigation decisions.

[04] Further, many modern irrigation systems depend heavily on renewable energy sources such as solar power. While sustainable, these systems suffer from reliability issues during low sunlight conditions or battery depletion. Existing solutions do not adequately integrate fallback mechanisms utilising available utility power fallback supply.

[05] Additionally, current systems do not effectively combine environmental sensing, uncertainty evaluation, and energy-aware decision-making into a unified framework for irrigation control.

[06] Accordingly, there exists a need for an intelligent irrigation system capable of integrating environmental data, evaluating forecast uncertainty, and dynamically managing energy sources, including fallback to utility electricity.

OBJECTIVES OF THE PRESENT INVENTION

[07] The primary objective of the present invention is to provide a cloud-based intelligent irrigation system for optimising water usage.

[08] Another objective is to provide a system capable of evaluating uncertainty in weather forecasts before making irrigation decisions.

[09] Another objective is to provide an adaptive irrigation strategy based on confidence levels derived from forecast analysis.

[10] Another objective is to provide an energy-aware irrigation control system utilising solar power and battery storage.

[011] Another objective is to provide a reliable fallback mechanism using the government's electricity supply when renewable energy sources are insufficient.

[012] Another objective is to provide a system that reduces water wastage while ensuring adequate irrigation.

[013] Another objective is to provide a scalable and remotely manageable irrigation system using cloud computing.

SUMMARY OF THE INVENTION

[014] The present invention provides a cloud-based irrigation decision system comprising environmental sensing modules, a communication interface, a cloud processing platform, and an irrigation control system.

[015] The system collects real-time data, including soil moisture, rainfall, temperature, and humidity, and transmits the data to a cloud platform for analysis.

[016] The cloud platform calculates crop water requirements based on evapotranspiration and soil water balance, and integrates weather forecast data.

[017] An uncertainty evaluation module determines a confidence level based on forecast probability, historical accuracy, and multi-source validation.

[018] An energy evaluation module assesses solar power availability and battery status, and selectively utilises the utility power fallback supply when renewable energy is insufficient.

[019] Based on the confidence level and energy status, the system determines whether to delay, reduce, or fully execute irrigation.

[020] The system then controls irrigation hardware, including pumps and valves, to deliver water efficiently.

BRIEF DESCRIPTION OF THE ACCOMPANYING DRAWINGS

[021] **Figure. 1** illustrates a flow diagram representing the intelligent irrigation decision-making process, including uncertainty evaluation and energy-based control.

[022] **Figure. 2** illustrates a system architecture of the cloud-based irrigation system, including sensing modules, a communication network, a cloud processing platform, and an irrigation control system with utility power fallback integration.

DETAILED DESCRIPTION OF THE INVENTION

[023] The intelligent irrigation decision process commences with initiation of an irrigation management cycle (100). Field data is collected (101) from a plurality of sensing devices deployed throughout an agricultural field. The collected data may comprise soil moisture content, rainfall measurements, ambient temperature, relative humidity, solar radiation levels, wind conditions, and other agronomic parameters associated with crop growth and irrigation planning.

[024] The collected field data is transmitted (102) through a communication network to a cloud processing platform for storage, processing, and decision analysis. The communication network may comprise GSM, LoRaWAN, NB-IoT, Wi-Fi, cellular communication, satellite communication, or combinations thereof.

[025] The cloud processing platform determines crop water requirements (103) using evapotranspiration calculations, soil water balance models, crop coefficient values, crop growth stage information, historical irrigation records, and current soil moisture measurements. The calculated water requirement represents the quantity of water necessary to maintain optimal crop growth conditions and serves as a baseline irrigation requirement.

[026] Weather forecast information is subsequently acquired (104) from one or more meteorological data providers. The acquired forecast information may include rainfall probability, expected rainfall quantity, forecast duration, humidity forecasts,

temperature forecasts, cloud cover information, and wind conditions relevant to irrigation scheduling.

[027] The forecast information is subjected to an uncertainty evaluation process (105), wherein forecast probability, historical forecast accuracy, multi-source weather validation, forecast variance, and confidence intervals are analysed to determine the reliability of the forecast data. Unlike conventional irrigation systems that directly rely upon weather forecasts, the present invention evaluates forecast reliability before making irrigation decisions.

[028] Based on the uncertainty evaluation, a confidence level decision (106) is generated. The confidence level represents the likelihood that the forecasted rainfall will occur within the predicted period and in a quantity sufficient to influence irrigation planning.

[029] When the confidence level exceeds a predetermined upper threshold, the forecast is classified as highly reliable, indicating a high probability of substantial rainfall. Under such conditions, irrigation is delayed (107) because the anticipated rainfall is expected to satisfy all or a substantial portion of the crop water requirement. Delaying irrigation prevents unnecessary water application, reduces energy consumption, and minimizes the risk of over-irrigation. During the delay period, updated weather forecasts and field conditions may continue to be monitored to determine whether irrigation should remain postponed.

[030] When the confidence level falls below a predetermined lower threshold, the forecast is classified as unreliable or insufficient for irrigation planning. Such conditions may occur when rainfall probability is low, forecast sources provide conflicting predictions, or historical prediction accuracy is inadequate. Under these circumstances, full irrigation execution (108) is selected, ensuring that the complete calculated crop water requirement is supplied irrespective of anticipated rainfall. This approach protects crops against moisture stress that could result from inaccurate weather forecasts.

[031] When the confidence level lies between the upper and lower thresholds, the forecast is classified as moderately reliable. Under such conditions, a reduced irrigation schedule (109) is selected. The irrigation volume may be proportionally reduced according to the confidence score, thereby partially accounting for anticipated rainfall

while ensuring adequate soil moisture availability. This strategy balances water conservation with irrigation reliability.

[032] Following selection of the irrigation strategy, an energy adjustment decision (110) is performed. The energy adjustment decision evaluates available solar energy generation, battery charge status, expected energy consumption, irrigation duration requirements, and utility electrical supply availability.

[033] When sufficient renewable energy resources are available, the selected irrigation strategy is executed without modification. However, when renewable energy availability falls below a predetermined threshold, adaptive energy management actions (111) are initiated. Such actions may include reducing irrigation duration, postponing irrigation, prioritizing critical irrigation zones, utilizing stored battery energy, or automatically switching to utility electrical supply. The utility power fallback mechanism ensures uninterrupted irrigation operation during periods of insufficient solar generation or battery depletion.

[034] The finalized irrigation decision is executed (112) through generation of control commands for pumps, valves, sprinklers, drip irrigation controllers, and associated irrigation hardware. The control commands regulate irrigation timing, flow rate, duration, and delivery volume according to the determined irrigation schedule.

[035] Upon completion of irrigation delivery, the irrigation cycle terminates (113). The process may subsequently restart according to a predefined schedule or upon receipt of updated environmental data requiring reassessment of irrigation requirements. Consequently, the method integrates environmental sensing, forecast reliability assessment, confidence-based irrigation scheduling, and adaptive energy management into a unified irrigation decision framework capable of reducing water wastage and improving irrigation efficiency.

[036] The cloud-based intelligent irrigation system (201) comprises an environment and soil monitoring module (202), an irrigation terminal (203), a communication module (204), a cloud processing platform (205), and an irrigation control system (211) configured to collectively perform uncertainty-aware and energy-adaptive irrigation management.

[037] The environment and soil monitoring module (202) serves as the primary sensing layer of the system and is configured to continuously monitor field conditions. The monitoring module may comprise soil moisture sensors, rainfall sensors, temperature sensors, humidity sensors, solar radiation sensors, wind sensors, and other environmental sensing devices. The collected sensor information provides a real-time representation of field conditions and crop water status.

[038] The sensed data is received by the irrigation terminal (203), which functions as a field-level control and data acquisition unit. The irrigation terminal may comprise a microcontroller, processor, memory unit, sensor interface circuitry, local storage, and control circuitry. The irrigation terminal performs signal conditioning, preliminary data validation, local data aggregation, and temporary storage before forwarding the collected information for cloud-based processing.

[039] The communication module (204) establishes bidirectional communication between field devices and the cloud processing platform (205). The communication module may support GSM, 4G/5G cellular communication, Wi-Fi, LoRaWAN, NB-IoT, satellite communication, or combinations thereof. Through the communication module, field data is transmitted to the cloud platform and irrigation control commands are transmitted back to the field.

[040] The cloud processing platform (205) functions as the central intelligence layer of the system and performs large-scale data processing, irrigation analytics, weather analysis, and decision generation. The platform receives field data from the irrigation terminal (203) and stores the information within a data storage and processing module (206).

[041] The data storage and processing module (206) maintains historical sensor records, weather data, crop information, irrigation schedules, and energy consumption records. The module further performs evapotranspiration calculations, soil water balance computations, moisture deficit analysis, and crop water requirement estimation. Based on these computations, the module determines the baseline irrigation requirement necessary to sustain optimal crop growth.

[042] The calculated irrigation requirement is supplied to an irrigation decision module (207). The irrigation decision module generates preliminary irrigation schedules by determining irrigation timing, irrigation duration, irrigation quantity, and irrigation priority levels. The generated schedule represents the amount of irrigation that would ordinarily be supplied under existing field conditions.

[043] The irrigation decision module (207) operates in conjunction with an uncertainty module (208), which evaluates the reliability of weather forecast information. The uncertainty module may analyse rainfall probability, forecast variance, historical prediction accuracy, forecast consistency among multiple weather service providers, and confidence intervals associated with predicted rainfall events. Based on these analyses, the uncertainty module generates a confidence score representing the likelihood of forecast accuracy.

[044] The generated confidence score is supplied to the irrigation decision module (207), enabling the irrigation schedule to be dynamically modified. When forecast confidence is high, the irrigation decision module may delay irrigation. When forecast confidence is low, full irrigation may be maintained. When forecast confidence is moderate, irrigation volume may be proportionally reduced. Consequently, irrigation decisions are influenced not only by crop water requirements but also by forecast reliability.

[045] An energy management module (209) continuously evaluates available energy resources required for irrigation execution. The energy management module monitors solar panel output, battery state-of-charge, energy consumption patterns, anticipated irrigation energy demand, and utility electrical supply availability. Based on these parameters, the energy management module determines whether sufficient renewable energy is available for irrigation execution.

[046] When renewable energy resources are adequate, irrigation operations proceed using available solar power and stored battery energy. When renewable energy resources become insufficient, the energy management module initiates adaptive energy management actions including irrigation rescheduling, irrigation duration adjustment, battery utilization, or automatic switching to utility electrical supply. The utility power fallback mechanism ensures reliable irrigation operation irrespective of renewable energy fluctuations.

[047] Outputs generated by the irrigation decision module (207), uncertainty module (208), and energy management module (209) are integrated within a decision output module (210). The decision output module performs decision fusion and generates final executable irrigation commands representing the optimized irrigation strategy.

[048] The generated irrigation commands are transmitted to the irrigation control system (211), which comprises water pumps, motor drivers, solenoid valves, sprinkler systems, drip irrigation controllers, flow regulation devices, and associated irrigation hardware. The irrigation control system executes the received commands by regulating water delivery according to the optimized irrigation schedule.

[049] Water is ultimately supplied to the agricultural field through a water delivery output (212). The delivered irrigation volume, duration, and timing correspond to the optimized irrigation strategy generated through combined environmental sensing, uncertainty analysis, and energy management. Accordingly, the system architecture provides a fully integrated cloud-based irrigation platform capable of improving irrigation accuracy, reducing water wastage, enhancing energy utilization, and ensuring reliable irrigation operation under changing environmental and energy conditions.

5. CLAIMS

We claim:

1. A cloud-based intelligent irrigation decision system comprising:

an environment and soil monitoring module (202) configured to collect field data including soil moisture, rainfall, temperature and humidity data;

an irrigation terminal (203) operatively connected to the monitoring module (202); a communication module (204) configured to transmit the field data to a cloud processing platform (205);

the cloud processing platform (205) comprising:

a data storage and processing module (206) configured to calculate crop water requirement;

an irrigation decision module (207) configured to generate irrigation control commands;

an uncertainty module (208) configured to determine forecast confidence based on weather uncertainty analysis; and

an energy management module (209) configured to assess renewable energy availability and selectively switch to utility power fallback;

wherein outputs of the irrigation decision module (207), uncertainty module (208), and energy management module (209) are combined to generate irrigation control commands for an irrigation control system (211) for adaptive irrigation execution.

2. A method for intelligent irrigation control comprising:

collecting field data;

transmitting the field data to a cloud platform;

calculating crop water requirement;

acquiring weather forecast information;

performing uncertainty evaluation on the weather forecast information;

determining a forecast confidence level;

generating irrigation control commands including delayed irrigation, reduced irrigation or full irrigation;

performing energy availability evaluation including utility power fallback determination; and

executing irrigation through pumps and valves based on the generated irrigation control commands.

3. The system as claimed in claim 1, wherein the uncertainty module (208) evaluates forecast confidence using forecast probability, historical prediction accuracy and multi-source forecast validation.

4. The system as claimed in claim 1, wherein the energy management module (209) is configured to switch between solar power, battery storage and utility electrical supply based on available energy levels.

5. The system as claimed in claim 1, wherein the irrigation decision module (207) generates reduced irrigation or delayed irrigation commands based on forecast confidence levels.

6. The system as claimed in claim 1, wherein the irrigation control system (211) comprises pumps, valves and irrigation actuators controlled through the generated irrigation control commands.

7. The system as claimed in claim 1, wherein the data storage and processing module (206) calculates water requirement using evapotranspiration and soil water balance models.

8. The system as claimed in claim 1, wherein the energy management module (209) modifies irrigation duration or switches to utility power fallback when renewable power falls below a threshold.

9. The method as claimed in claim 2, wherein uncertainty evaluation includes assigning a confidence score to predicted rainfall and selecting irrigation execution based on the assigned confidence score.

10. The method as claimed in claim 2, wherein irrigation execution is adjusted jointly according to forecast confidence level and energy availability.



6. DATE AND SIGNATURE

Dated this 27th day of May, 2026

Vinoth Kumar J
Patent Agent (IN/PA 6111)
Agent for the Applicant

7. ABSTRACT OF THE INVENTION

A CLOUD-BASED INTELLIGENT IRRIGATION DECISION SYSTEM WITH UNCERTAINTY-AWARE CONTROL AND UTILITY POWER FALLBACK

A cloud-based intelligent irrigation decision system and method are disclosed for adaptive irrigation control using uncertainty-aware decision making and utility power fallback. The system comprises an environment and soil monitoring module (202) configured to collect field data including soil moisture, rainfall, temperature and humidity, an irrigation terminal (203), a communication module (204), and a cloud processing platform (205). The cloud processing platform includes a data storage and processing module (206), an irrigation decision module (207), an uncertainty module (208), and an energy management module (209). Based on crop water requirement calculations, forecast confidence evaluation and energy availability assessment, irrigation control commands are generated through decision output (210) for controlling an irrigation control system (211) to deliver water through output (212). The associated method includes collecting field data (101), uncertainty evaluation (105), confidence-based irrigation decision (106–109), energy adjustment (110–111), and irrigation execution (112). The invention improves irrigation reliability, reduces water wastage and enables energy-adaptive irrigation management.